



COMPUTER SCIENCE 180A (FALL, 2025) ALGORITHMS

PROBLEM SET 2

- Students are required to work in pair. Working in pairs entails jointly working in person on the entire assignment
- One submission for group
- Name your PDF file LastFirst_LastFirst-PS2.pdf (e.g., TimSmith_ClaireChair-PS2.pdf)
- If you consult an outside resource (e.g., web page, book, or research paper) to arrive at your solution, be sure to cite that resource
- You may not discuss the homework problems with students outside of your group
- All problem sets must be typeset and turned in as a PDF file through Moodle the date it is due.
- Whenever possible please limit your answer to each problem to a single page. When this is not possible, a page break should be inserted between problems.
- No handwritten assignments will be graded.
- No late submissions will be accepted.

Q1. Recall the interval scheduling problem discussed in class, which is an example of an activity-selection problem.

Suppose that instead of always selecting the first activity to finish, you instead select the last activity to start that is compatible with all previously selected activities. Describe how this approach is a greedy algorithm and prove that it yields an optimal solution.

Q2. Consider a modification to the activity-selection problem in which each activity a_i has, in addition to a start and finish time, a value v_i . The objective is no longer to maximize the number of activities scheduled, but instead to maximize the total value of the activities scheduled. That is, the goal is to choose a set A of compatible activities such that $\sum_{a_k \in A} v_k$ is maximized. Give a polynomial-time algorithm for this problem.

Q3. Recall the 0-1 knapsack problem discussed in class. A variation of this is the fractional knapsack problem, where the setup is the same, but the thief is allowed to take fractions of items instead of making an all-or-nothing (0-1) choice for each item.

Prove that the fractional knapsack problem has the greedy-choice property.

Q4. What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers?

a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21

Can you generalize your answer to find the optimal code when the frequencies are the first n Fibonacci numbers?

Q5. Generalize Huffman's algorithm to ternary codewords (i.e., codewords using the symbols 0, 1, and 2), and prove that it yields optimal ternary codes.

Q6. Give a simple implementation of Prim's algorithm that runs in $O(V^2)$ time when the graph $G = (V, E)$ is represented as an adjacency matrix.

Q7. Suppose that all edge weights in a graph are integers in the range from 1 to $|V|$. How fast can you make Kruskal's algorithm run? What if the edge weights are integers in the range from 1 to W for some constant W ?

Q8. A *bottleneck spanning tree* T of an undirected graph G is a spanning tree of G whose largest edge weight is minimum over all spanning trees of G . The value of the bottleneck spanning tree is the weight of the maximum-weight edge in T .

- a. Argue that a minimum spanning tree is a bottleneck spanning tree. That is, finding a bottleneck spanning tree is no harder than finding a minimum spanning tree.
- b. Give a linear-time algorithm that, given a graph G and an integer b , determines whether the value of the bottleneck spanning tree is at most b .